

The state of Python in India

>>Python is popular in developer circles, primarily due to its adaption by companies such as Google among several others; and is now gaining momentum in India as well> by Noufal Ibrahim

Way back in 1991, Guido van Rossum developed what became known as Python, a programming language. It grew from a small project into a first class programming language that powers everything from modern websites to games to scientific computing and even NASA projects. Although initially used in prototyping and toy projects for tinkerers, it now commands attention from many companies that actually look for it on a resume.

The Python reference implementation is available as free software under a liberal licence and the overall direction of the language is guided by a community process. This lends itself to the creation of communities of enthusiasts that support and help in the development of the language. Being a country with a rich IT culture, many Python communities have grown and flourished in India. The language was designed with the notion that code has to be read more than written. Python code is English-like and has often been called “executable pseudo code”. This makes it ideal for newcomers as well as people in various fields that aren’t professional programmers.

Python’s success in the scientific / computing community is an example of this. The dynamic nature of the language enables quick turnarounds and increases programmer productivity. The huge standard library makes it very easy to get started with almost any kind of application. Add to all this a thriving 20-year old community of enthusiasts and developers and you have all the elements needed to make a successful, reliable and high quality open source project. Python is also extendable in C making it a very nice “glue language”. It is possible to write a little C extension for Python that can allow it to, for example, interface with a piece of hardware that can then be controlled from within the language console. Just like it is extendable, Python can be embedded. This means that Python can be used as a scripting interface for a larger application. Many graphical applications implement their performance intensive parts in C and provide a Python scripting interface that can be used to extend the application.

Emergence as a powerful language

While initially, it was only used by programmers for their personal projects, it has taken root in the industry and a quick search on popular job boards for Python displays numerous positions available. The short development time and dynamic nature of the language

is very attractive to startups and such companies for whom rapid turnarounds are important. Larger companies rely on Python for their internal infrastructure and some even ship applications written in Python to their customers.

Adaption by the industry

Google relies on Python heavily for many of its internal infrastructure projects and provides Google App Engine which is a PaaS product that supports three languages, one of which is Python. In fact, Google employs the creator of Python himself, as well as many other luminaries of the Python community. ZeOmega is another company with a long history of involvement with Python technically, as well as socially. They ship a number of products written in



PHOTO COURTESY: KUSHAL DAS

Get in touch with the Python community by joining the user group in your city

Python and have backed the community and its conferences for a long time now. Enthought, with its offices in Mumbai ships a suite of scientific computing products mostly written in Python.

Python in academia

Perhaps the most widespread commercial use of Python, in India anyway, is for writing web apps. There are a number of companies, large and small, in India who have public facing and internal web-based tools that are written using one of the many Python web frameworks out there. On the academic side, there’s been an increasing interest in using Python as a first language for new programmers. Various engineering colleges in Bangalore such as MSRIT and R.V.